

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY  
4<sup>th</sup> Semester of Bachelor of Technology (CE)

University Examination, May 2011

CE208: Object Oriented Concept and Programming using VC++

Date: 11/05/2011

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 70

**Instructions:**

1. Figure to the right indicates full marks
2. Make suitable assumption whenever necessary and mention it clearly
3. Parts of same question should be answered together and in the same sequence.
4. Answers to the sections must be written separately.

**SECTION-I**

**Q-1 Answer the followings:**

1. Define terms with example: Class, Object. [02]
2. When will you make function inline? Why? [02]
3. Explain concept of friend function with example. [02]
4. Can we use the same function name for a member function of a class and an outside function in the same program file? If yes, how are they distinguished? If no, give reason. [02]
5. What is constructor? Explain explicit and implicit call/invoke of constructor with example. [03]

**Q-2 Attempt the following:**

1. What is purpose of inheritance? We know that a private member of a base class is not inheritable. Is it anyway possible for the objects of a derived class to access the private members of the base class? If yes, how? Remember, the base class cannot be modified. [12]
2. What is virtual function? Why do we need virtual function? List at least four basic rules for creating virtual functions which satisfy the compiler requirements.
3. Explain why a friend function can not be used to overload assignment operator =. Write a program to use overloaded == operator to compare two string. [12]

**OR**

**Q-2 Attempt the following:**

- [A] Create FLOAT class that contains one float data member. Overload all the four arithmetic operators so that they operate on the object of FLOAT class.
- [B] What does *this* pointer point to? What are the applications of *this* pointer? Explain using example.
- [C] In what order are the constructors called when a derived class object is created? Class D is derived from class B. The class D does not contain any data members of its own. Does the class D required constructor? If yes, why? If no, give justification.

**Q-3 Attempt the following questions:**

- [A] Explain essential components of Windows program structure with diagram. [04]
- [B] What is the difference between opening a file with a constructor function and opening file with *open()* function? [02]



- [C] What are the steps involved in using a file in a C++ program? Write a program that reads a text file and creates another file that is identical except that every sequence of consecutive (two or more) blank spaces is replace by a single space. [06]

OR

- [C] Explain the document / view concept in MFC. Also explain linking of a document and its views with graphical representation. [06]

## SECTION-II

### Q-4 Attempt following:

1. What do you mean by OOAD? What are the most important features of OOAD? [03]
2. What is the purpose of interaction diagrams? How they differ from static class diagram? [03]
3. What are swimlanes? It is related to which UML diagram? [02]
4. Explain terms "Extends" and "Include" relationship in Usecase diagram. [02]
5. When do we declare a class as abstract? [01]

### Q-5 Attempt following:

- [A] What are the advantages of function prototypes in C++? What is the main advantage of passing arguments by reference? [05]
- [B] When do we declare a member of a class static? List two main properties of static member function. [02]
- [C] What are input and output stream? Describe the various classes available for file operation? [05]

OR

- [C] What are the different forms of inheritance? Explain each with example. [05]

### Q-6 Attempt any two:

- [A] How a **state chart** diagram can be used to model the object? Explain symbols of state chart with example. [03]
- [B] Differentiate:  
(a) Event and Action.  
(b) State and Activity. [04]
- [C] Draw the Class diagram for ATM [05]

OR

- Draw the Interaction diagram for ATM [05]